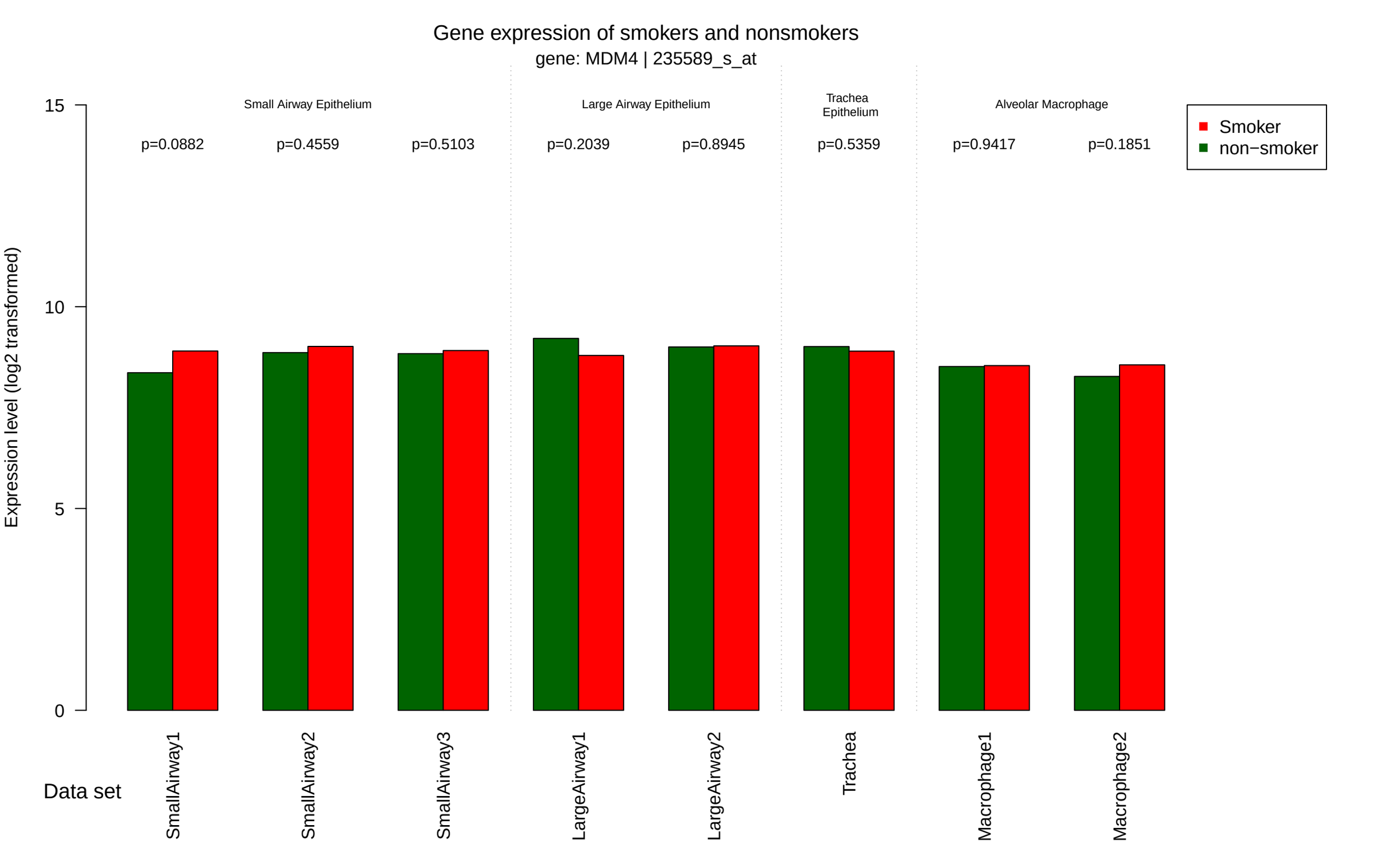
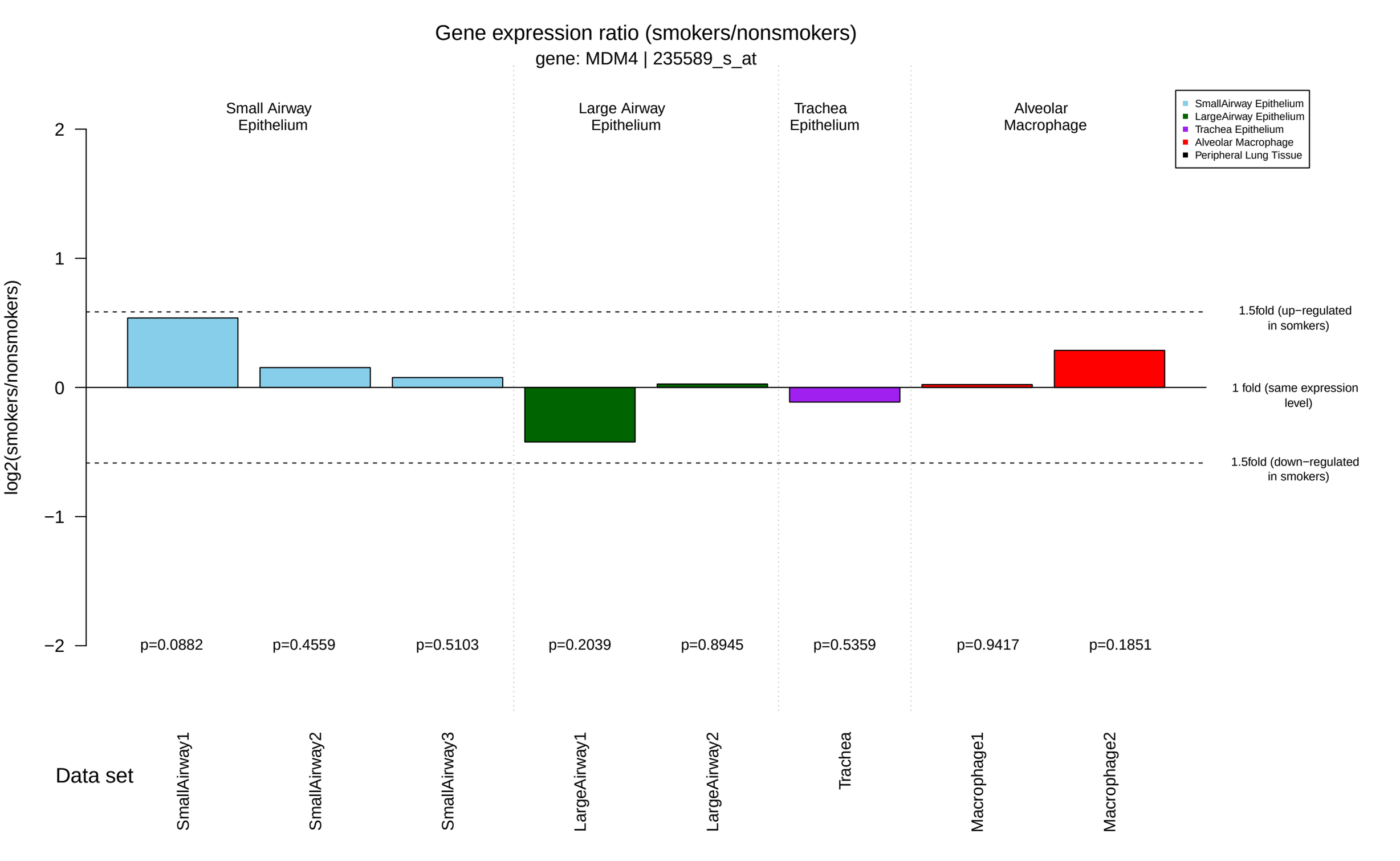
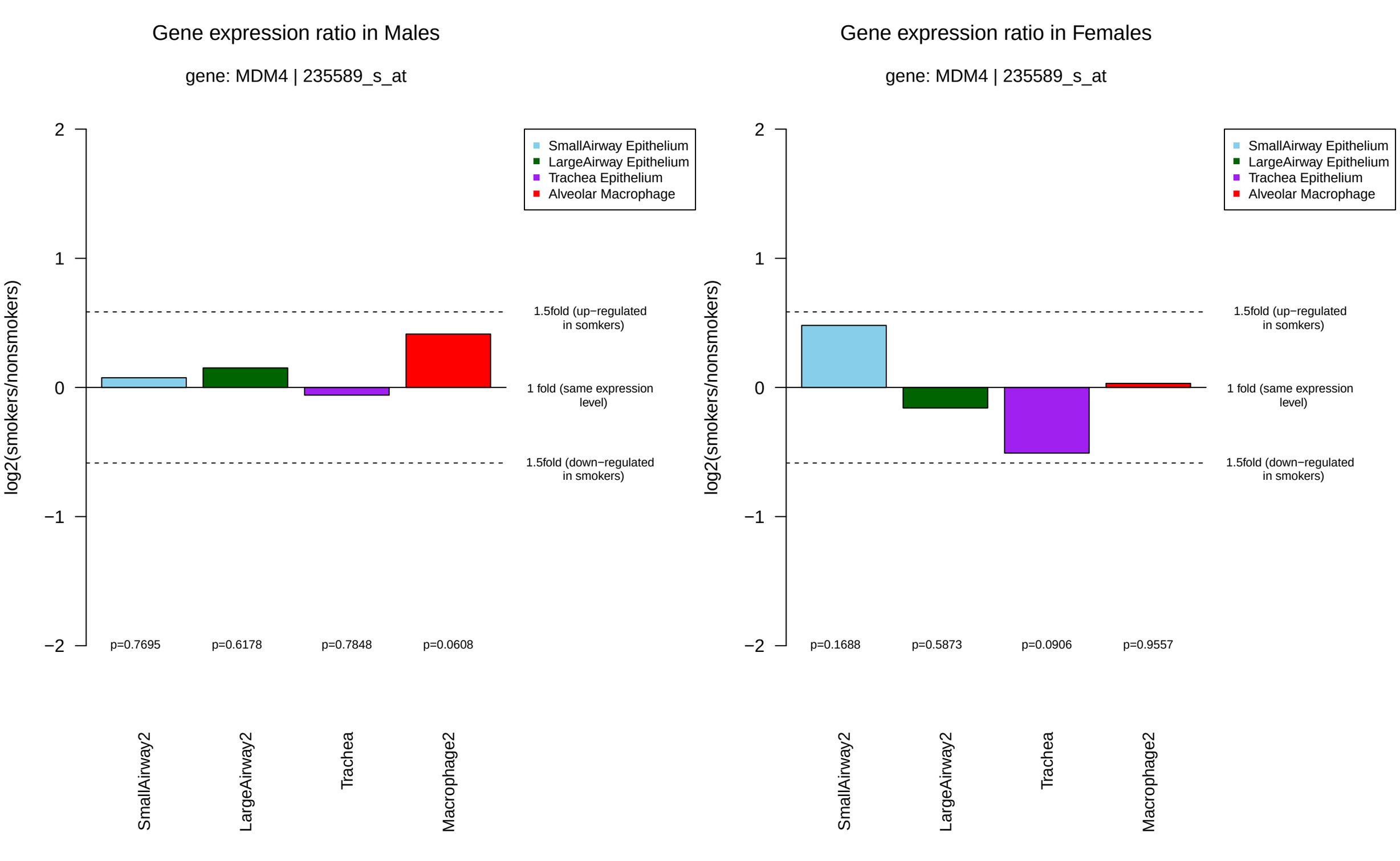
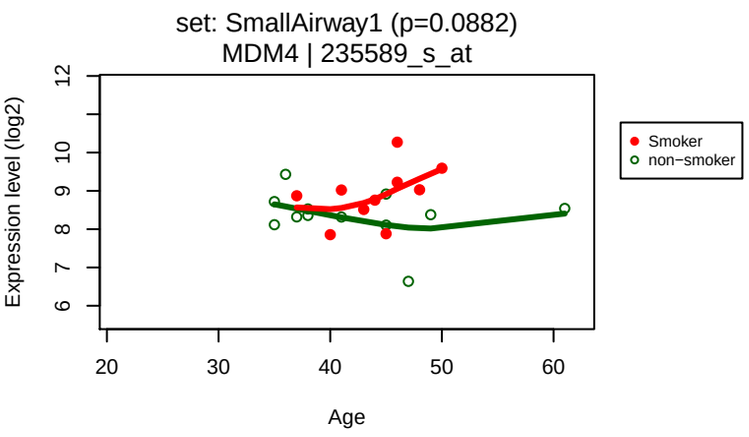
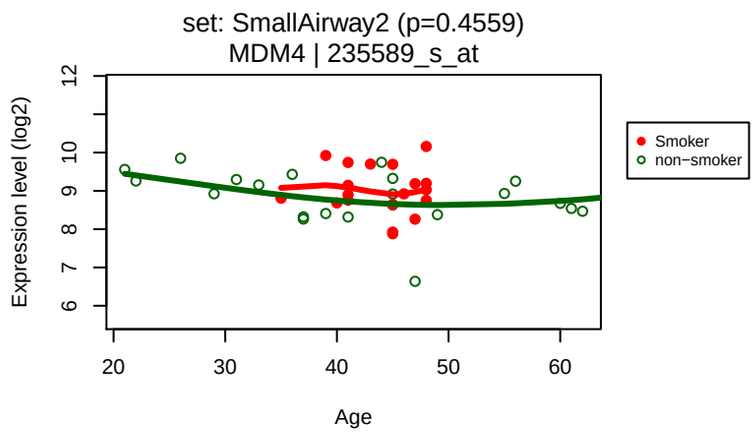
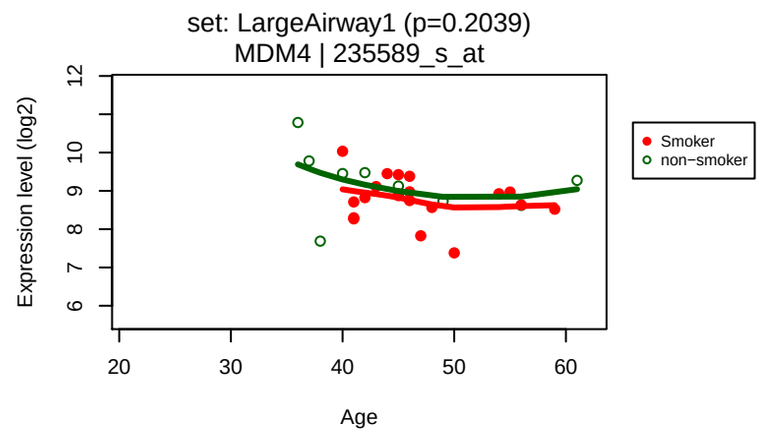

Gene expression of smokers and nonsmokers
gene: MDM4 | 235589_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0882
p=0.4559
p=0.5103
p=0.2039
p=0.8945
p=0.5359
p=0.9417
p=0.1851
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

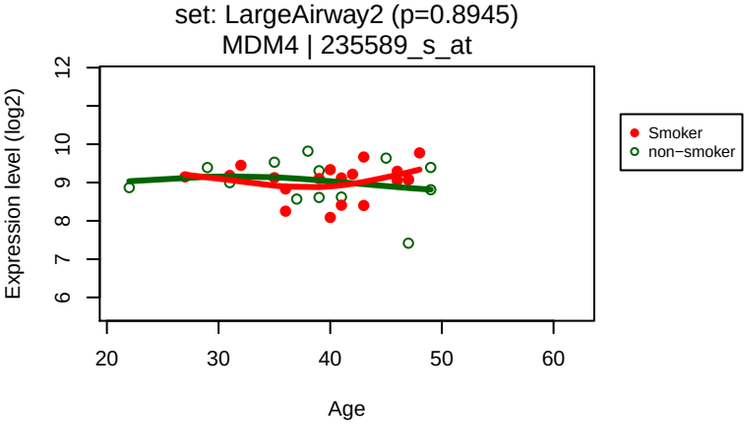

Gene expression ratio (smokers/nonsmokers)
gene: MDM4 | 235589_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0882
p=0.4559
p=0.5103
p=0.2039
p=0.8945
p=0.5359
p=0.9417
p=0.1851
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

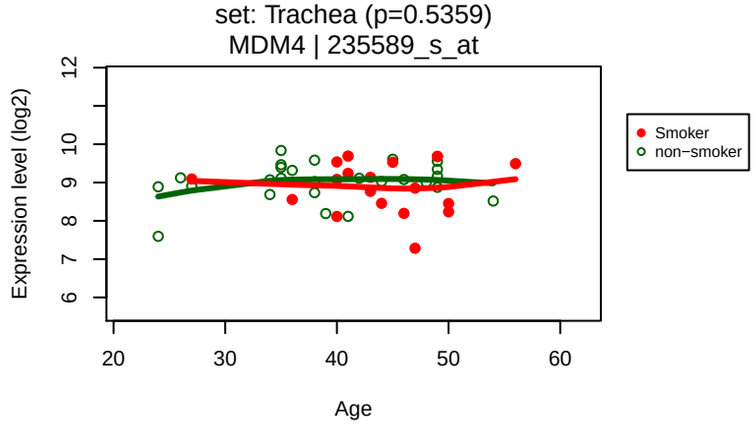

Gene expression ratio in Males
gene: MDM4 | 235589_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7695
p=0.6178
p=0.7848
p=0.0608
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MDM4 | 235589_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1688
p=0.5873
p=0.0906
p=0.9557
SmallAirway2
LargeAirway2
Trachea
Macrophage2

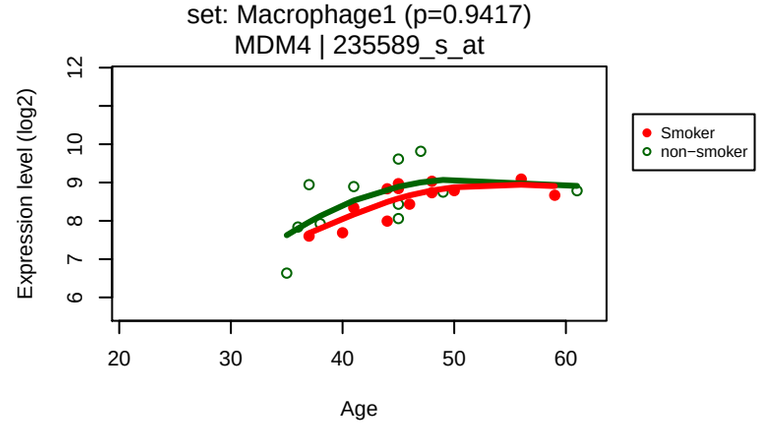

set: SmallAirway1 (p=0.0882)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker

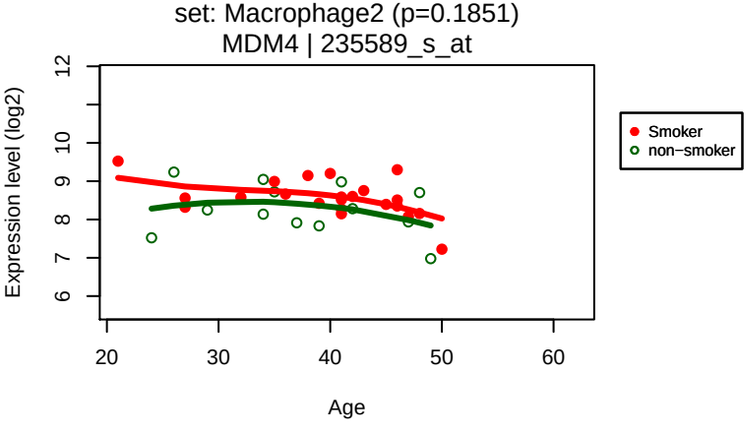

set: SmallAirway2 (p=0.4559)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker

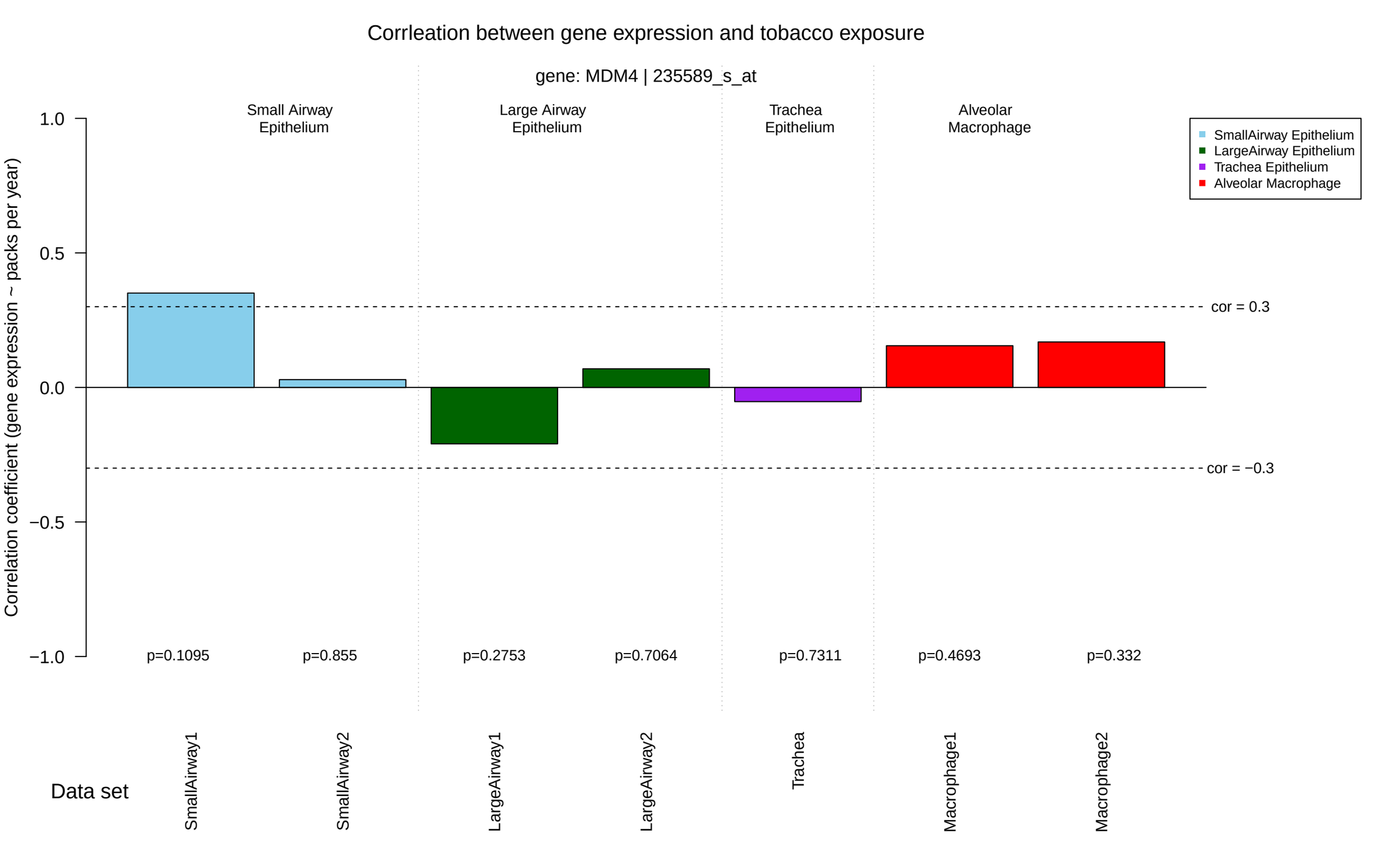

set: LargeAirway1 (p=0.2039)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8945)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5359)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9417)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1851)
MDM4 | 235589_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MDM4 | 235589_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1095
p=0.855
p=0.2753
p=0.7064
p=0.7311
p=0.4693
p=0.332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2